A Digital-Twin Multi-Agent Framework for Enterprise Cognitive Networking: ECNetBench and Leakage-Safe Multi-Seed Evaluation

Anonymous Author(s) Affiliation withheld for double-blind review
Email: withheld@example.org

***Abstract*—Enterprise cognitive networking lacks a shared, leakage-safe benchmark that couples multi-layer configuration, multi-modal telemetry, and supervised NetOps labels. We present *ECNetBench*, a synthetic enterprise benchmark under AOS-CX-style operational assumptions, and an *Enterprise Cognitive Network* (ECN-v3) with Perception, Anomaly, Prediction, RCA, Impact, and Healing agents. The final T1 detector combines leakage-safe temporal enrichment with anchored telemetry-first fusion; nesting-safe stacking is retained only as a negative ablation. On six frozen instances with a temporal $70/15/15$ protocol, ECN-v3 attains mean T1 AUPRC 0.115 (bootstrap 95% CI $[0.068, 0.168]$), above random forest (0.076) and the prior v2 configuration (0.058), while paired tests versus RF remain non-significant at $n{=}6$. The recommended T2 head is telemetry logistic regression. The contribution is a reproducible multi-seed benchmark and an evidence-selected detection–RCA–healing decision-support pipeline with honest multi-seed statistics.**

***Index Terms*—Enterprise networking, digital twin, multi-agent systems, anomaly detection, failure prediction, root-cause analysis, AIOps, benchmark, reproducibility.**

I. Introduction

Modern enterprise campuses and data-center fabrics are operated under continuous change: VLAN and ACL updates, EVPN/VXLAN overlays, QoS policy edits, and firmware-driven control-plane behavior. Cognitive NetOps and AIOps promise earlier anomaly detection, failure-horizon prediction, explainable RCA, service-impact estimation, and remediation recommendations [1]–[3]. Beyond purely technical detectors, enterprise modernization also depends on governance maturity, workforce adaptation, and cybersecurity readiness when operators adopt digital twins and automation [4]–[7]. In practice, progress is slowed by fragmented public resources: topology archives without multi-modal telemetry [8], security flow corpora that omit enterprise L2/L3 configuration semantics [9], and telemetry/RCA artifacts that do not jointly expose recovery and impact labels under a temporal leakage protocol.

This paper addresses that gap with two coupled contributions. First, **ECNetBench** is a synthetic enterprise cognitive-networking benchmark whose schema, failure taxonomy, and label definitions are designed for leakage-safe supervised learning under AOS-CX-style operational assumptions (inventory, VLAN/ACL/QoS, routing, counters, syslog/alerts, NAE-like analytics, and recovery/impact fields). Second, we implement and evaluate an **Enterprise Cognitive Network (ECN-v3)**: a Digital Twin graph plus a multi-agent stack (Perception, Anomaly, Prediction, RCA, Impact, Healing) with leakage-safe feature enrichment, *anchored* telemetry-first fusion, and TreeSHAP-explained RCA. The novelty is the coupled benchmark, explicit temporal leakage protocol, and evidence-selected fusion evaluation on six frozen instances—not a claim of absolute historical priority over all related AIOps or digital-twin systems.

Our contributions are: (i) ECNetBench design (relational/graph schema, causal incident modeling, leakage-safe labels, and six frozen instances); (ii) the ECN-v3 framework with architecture isolation of anchored fusion versus stacking; (iii) multi-seed evaluation with baselines, ablations, calibration, confidence intervals, and paired significance tests under $n{=}6$; and (iv) a reproducible package with checksums and relative-path evaluation. Empirically, the final anchored T1 head improves mean AUPRC relative to the prior v2 configuration and strong tabular baselines, while paired tests versus random forest remain non-significant and stacking is a negative ablation; detailed metrics appear in Section VI.

The overall architecture is introduced in Fig. 2 (Section IV). Section II situates related work; Sections III–IV detail the benchmark and framework; Sections V–VI present protocol and results; Sections VII–IX discuss implications, limitations, and conclusions.

II. Related Work

A. Network digital twins and autonomous networks

Network digital twins (NDTs) aim to provide data-driven, near-real-time models for what-if analysis and closed-loop control [3], [10]. Standards bodies and industry white papers discuss NDTs as enablers of autonomous networks and safe change validation [11], [12]. Our twin focuses on enterprise campus/DC inventory and telemetry graphs rather than packet-level WAN simulation, and is evaluated as a feature provider inside a multi-agent NetOps pipeline.

B. AIOps, anomaly detection, and RCA

Surveys of AIOps and network anomaly detection highlight class imbalance, concept drift, and the need for operationally meaningful metrics beyond accuracy [1], [2], [13]. Classical detectors (Isolation Forest [14], EWMA) and strong tabular learners remain competitive when features are informative. RCA research spans log mining, causal graphs, and Clos-fabric diagnostics [2]. We evaluate explainable category-level RCA with ablations that remove twin or syslog cues, and we treat healing as *decision support* rather than live switch actuation.

C. Graph learning for networks

Graph neural networks and message-passing models have been proposed for topology-aware prediction [15], [16]. We evaluate both a historical GraphSAGE-style *proxy* (LightGBM on a restricted feature subset; retained for continuity) and a **true** GraphSAGE mean-aggregator over the Digital Twin device graph. Attention-tabular models such as TabNet [17] are included as a modern deep tabular baseline. The proposed ECN uses an explicit Digital Twin with neighbor aggregates and anchored fusion rather than claiming end-to-end GNN or TabNet superiority.

D. Benchmarks and datasets

Public resources include Topology Zoo [8], SNDlib [18], and security datasets such as UNSW-NB15 [9]. Enterprise cognitive NetOps additionally needs configuration semantics, multi-modal telemetry, service impact, and recovery labels under temporal freeze rules. ECNetBench is synthetic by design; we therefore emphasize leakage audits, seed robustness, and explicit threats to external validity (Section VIII).

E. Intent-based and multi-agent network management

Intent-based networking and multi-agent orchestration are long-standing themes in automated management [19], [20]. Recent surveys of agentic and multi-agent systems emphasize tool use, evaluation benchmarks, and governance frameworks for delegated autonomy [21], [22]. ECN-v3 instantiates specialized agents with an orchestrator; fusion is constrained (anchored telemetry weight ≥ 0.5) after architecture isolation showed nesting-safe stacking does not improve mean T1 AUPRC on ECNetBench. RCA explanations use TreeSHAP on a random-forest category model.

F. Governance, cybersecurity, and operational trust

As NetOps pipelines become more automated, enterprise deployments inherit broader digital-transformation and cybersecurity concerns: risk controls in IT organizations [23], trust-oriented auditing of operational records [24], and zero-exposure practices for identity-recovery workflows [25]. Parallel literature on enterprise generative and agentic AI highlights security/governance risks and prioritization architectures for retrieval-augmented or generative assistants [26], [27]. These works motivate treating ECN healing outputs as governed decision support rather than unsupervised actuation, and motivate leakage-safe evaluation and artifact traceability as scientific analogues of operational auditability.

III. ECNetBench: Benchmark Design

A. Design goals

ECNetBench targets enterprise cognitive networking research with five goals: (G1) AOS-CX-style multi-layer configuration realism; (G2) multi-modal telemetry (counters, resources, syslog/alerts, flow aggregates); (G3) causal incident chains linking failure, detection, recovery, and impact; (G4) leakage-safe supervised labels; (G5) multi-seed robustness under frozen instances. Table I contrasts ECNetBench with common public resource classes for cognitive NetOps research.

TABLE I
Qualitative comparison of ECNetBench with common public NetOps resource classes.

Resource class	Config.	Telemetry	RCA/heal	Multi-seed
Topology archives	Partial	No	No	Rare
Security flow sets	Low	Flows	Limited	Varies
Clos/telemetry RCA	Fabric-spec.	Yes	RCA-focused	Varies
ECNetBench	**Yes**	**Yes**	**Yes**	**Yes**

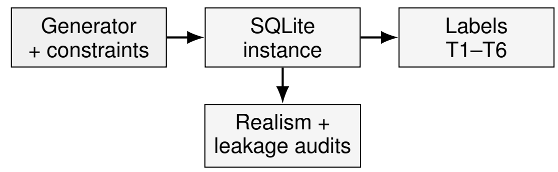

Fig. 1. ECNetBench generation and validation workflow: constrained synthesis produces a frozen SQLite instance that is audited for realism/leakage and exported to supervised task labels.

B. Enterprise assumptions and topology

Instances model a compact enterprise fabric (19 devices, 31 links in the evaluated family) spanning campus/DC roles with VLAN membership, ACL/QoS bindings, routing (e.g., BGP/OSPF-related tables), LAG/VSX-like redundancy concepts, and overlay-related objects where applicable. Topology snapshots and typed graph edges support Digital Twin construction. Generation constraints and schema documents accompany the release.

C. Telemetry and causal incident modeling

Synthetic generators emit interface counters, device resources, environmental/power samples, syslog/alert streams, and service KPIs. Failure incidents are drawn from a taxonomy covering link, device, control-plane, policy, and service-impacting families. Each incident carries detection, recovery-action, and impact fields so that labels for anomaly windows, failure horizons, RCA categories, impact, and healing actions remain consistent with a causal narrative. Fig. 1 summarizes the generation and validation workflow from constrained synthesis to audited SQLite instances and supervised labels.

D. Leakage-safe labels and splits

Labels are aligned to time bins. Features at decision time t_0 use only information available at or before t_0 (historical shifts; no future counters). Evaluation uses a temporal $70/15/15$ train/validation/test split frozen per instance. Forbidden fields (raw future labels, oracle incident IDs in feature matrices) are audited in the companion leakage reports.

E. Instances and seed robustness

We evaluate six frozen SQLite instances: primary instance v1.1.0-INST and independent seeds 101, 202, 303, 404, and

TABLE II
FROZEN EVALUATION INSTANCES USED IN THIS STUDY.

Instance folder	Role	Devices / links
`v1` (v1.1.0-INST)	Primary frozen instance	19 / 31
`v1.1-seed101`	Independent seed	19 / 31
`v1.1-seed202`	Independent seed	19 / 31
`v1.1-seed303`	Independent seed	19 / 31
`v1.1-seed404`	Independent seed	19 / 31
`v1.1-seed505`	Independent seed	19 / 31

TABLE III
SUMMARY OF ECNETBENCH RELATIONAL SCHEMA FAMILIES.

Family	Representative contents
Inventory/topology	Devices, interfaces, topology snapshots, graph nodes/edges
Configuration	VLAN/ACL/QoS, routing processes, configuration snapshots/diffs
Telemetry	Interface counters, resources, syslog/alerts, flow aggregates
Services	Applications, services, SLA and impact bindings
Incidents/labels	Failure incidents, recovery actions, supervised label tables

505 (Table II). SHA-256 digests are published in the instance checksum manifest. Seed-robustness and publication-validation reports accompany the package. SQLite is authoritative for evaluation; CSV/Parquet exports are optional derivatives. Schema families are summarized in Table III.

IV. ENTERPRISE COGNITIVE NETWORK FRAMEWORK

A. Digital Twin

The Digital Twin materializes inventory and topology as a typed graph synchronized with telemetry windows. Twin-derived features include residual contrasts (e.g., device resource versus neighbor summaries), structural aggregates, and interface error/discard/carrier deltas constructed with causal shifts. Twin features are optional inputs to agents and are ablated in twin-off / twin-only settings.

B. Leakage-safe feature enrichment (ECN-v3)

Beyond legacy aggregates, ECN-v3 adds causal rolling/EMA statistics, second differences, error-burst accumulators, and neighbor-instability proxies. All rolling and expanding operators exclude the current bin via a one-step causal shift; labels join features at a decision time offset of $30\,\mathrm{min}$ before the labeled window start. Isolation studies attribute the T1 gain primarily to this enrichment rather than to fusion complexity.

C. Multi-agent architecture

Fig. 2 depicts the agent workflow. Perception builds leakage-safe feature matrices from telemetry and twin views. Anomaly and Prediction score T1 anomaly windows and T2 failure-horizon risk, respectively. RCA predicts failure category with TreeSHAP explanations (T3); Impact estimates

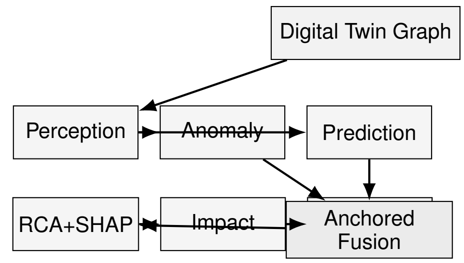

Fig. 2. ECN-v3 multi-agent workflow: Perception feeds Anomaly/Prediction specialists into *anchored* orchestrated fusion, followed by TreeSHAP RCA, impact estimation, and healing decision support.

service-impact labels (T4); and Healing recommends remediation classes as *decision support* (TR-AUTO). Actions are not executed on physical switches in this study. Architecture selection under the frozen six-seed protocol compares nesting-safe stacking against anchored fusion on the same enriched features. Anchored fusion constrains the telemetry specialist weight to be at least 0.5 (or selects telem-only), yielding higher mean T1 AUPRC with a simpler operator-facing interpretation. Stacking is retained as an ablation (negative result). The recommended T2 head is telemetry logistic regression under ultra-rare positives.

D. Class imbalance and calibration

Positive rates are low ($\sim$1% for T1; lower for T2). Training uses imbalance-aware weighting where applicable; Isolation Forest scores are mapped via a train empirical CDF. Thresholds for F1/precision/recall are tuned on validation only and are *not* used for AUPRC. Primary ranking metrics are AUPRC and ROC-AUC; Brier score reports calibration. Optional Beta/Platt calibration may be applied for probability display without changing AUPRC ranking.

V. EVALUATION PROTOCOL

A. Tasks

We report T1 anomaly detection, T2 failure-horizon prediction, T3 RCA (macro-F1 with TreeSHAP), T4 impact, and TR-AUTO healing decision support. Supplementary tasks (degradation, config risk) are produced by the harness and archived with the evaluation package.

B. Baselines and fairness

Baselines include logistic regression, random forest, LightGBM, XGBoost, CatBoost, gradient boosting, balanced random forest, Isolation Forest, EWMA, threshold rules, majority class, an MLP sequence proxy, **TabNet** [17] (telemetry-only), a historical GraphSAGE-style *proxy* implemented as LightGBM on a restricted feature subset, and a **true GraphSAGE** [15] message-passing model over the Digital Twin device adjacency (pure PyTorch mean aggregator; node features = telemetry + light twin scalars per device–time bin). Classical tabular baselines train on telemetry-only feature matrices; the proposed T1 model uses twin+telemetry with **anchored** fusion on leakage-safe enriched features. Shared temporal enrichment is applied consistently so comparisons are not confounded by unequal feature engineering for baselines versus proposed specialists. We explicitly distinguish the historical GNN *proxy* from the true GraphSAGE baseline; neither replaces the final ECN-v3 T1 head.

C. Metrics and statistics

The primary ranking metric is average precision (AUPRC). Secondary metrics include ROC-AUC, precision, recall, F1 (validation-tuned threshold), and Brier score. We report means and 95% confidence intervals across six seeds. Paired Wilcoxon signed-rank tests and Cliff's δ compare the final T1 head with each baseline and with the stacking ablation on per-seed AUPRC. With many paired comparisons and n=6, p-values are interpreted as exploratory evidence rather than family-wise controlled discoveries; we do not apply multiplicity correction and therefore do not claim simultaneous significance across the baseline suite.

D. Ablations

Primary architecture ablations contrast (i) v2 legacy features + anchored fusion, (ii) v3 enriched features + stacking, and (iii) v3 enriched features + anchored fusion (final). Module deltas report feature-enrichment gain versus v2, twin contribution under the final head, and stacking versus anchored on identical features.

E. Computational cost

Full-suite wall-clock time with TabNet and true GraphSAGE averages approximately 120 s per seed. Mean T1 training time for the anchored head remains $\approx$ 2.2 s; TabNet and GraphSAGE training are substantially slower ($\approx$ 16 s each on CPU).

VI. Results

Unless noted, multi-seed means use six frozen instances. Final T1 claims use the architecture-selection anchored head; baselines, deep models, and most ablations use the full-suite harness. Exact floating-point archives are listed in Appendix A.

A. T1 anomaly and T2 failure prediction

Table IV summarizes ranking metrics. The final ECN-v3 anchored detector achieves mean T1 AUPRC 0.115 (bootstrap 95% CI $[0.068, 0.168]$), above random forest (0.076) and the v2 legacy configuration (0.058), with mean T1 ROC-AUC 0.753. Fig. 3 shows the corresponding AUPRC means with 95% intervals for T1 and T2.

On T2, the recommended head is telemetry logistic regression (mean AUPRC 0.038), higher than RF (0.018). The ECN T2 columns in Table IV report the multi-agent proposed head (0.015), which is *not* the recommended operating head. We do not claim that multi-agent fusion improves T2 under ultra-rare positives.

Representative ROC and precision–recall curves appear in Fig. 4. Discrimination is moderate on ROC, while precision–recall remains challenging for all methods.

Paired Wilcoxon tests and Cliff's δ (Table V) show that the T1 mean advantage versus random forest is not significant at α=0.05 (p=0.688, δ=+0.389). Versus logistic and several weaker baselines, paired p-values reach 0.031 (exploratory under multiple comparisons). Stacking versus anchored is not significant (p=0.375).

B. Architecture selection and fusion ablation

Fig. 5 and Table VI isolate T1 sources of improvement. Leakage-safe feature enrichment accounts for the dominant gain; replacing anchored fusion with stacking on the same enriched features reduces mean AUPRC from 0.115 to 0.100. Stacking is therefore a negative ablation.

The T1 `full` ablation row is a full-suite re-run and may differ slightly from the Table IV selection mean. Table VII and Fig. 6 summarize contribution deltas. Feature enrichment versus v2 contributes $+0.058$ mean AUPRC. Twin gain under the final head is near zero. Stacking relative to anchored contributes -0.015. The harness module table reports a slightly negative twin ablation delta (-0.004); both estimates indicate Twin is not the primary T1 ranking driver.

C. Calibration and operating points

Table VIII summarizes calibration. Final T1 raw Brier is 0.151, worse than RF (0.023), motivating optional Beta calibration for probability display without changing AUPRC ranking. Fig. 7 shows a reliability diagram on v1.1.0-INST.

D. RCA, healing decision support, and cost

Table IX reports RCA macro-F1, healing scores, and cost. Proposed T3 RCA macro-F1 is 0.254 (RF+TreeSHAP). Healing with RCA category features reaches macro-F1 1.000 on this synthetic setup—an upper bound, not operational perfection; the honest no-category setting yields 0.190. Fig. 8 shows a representative confusion matrix. Full-suite evaluation averages about 119.7 s per seed.

E. Why simplicity remains preferable

At $\sim$1% positives, enriched telemetry recovers most ranking signal. Anchored fusion is simpler than stacking, improves mean T1 AUPRC on the same features, and remains competitive with RF in mean while acknowledging non-significant paired tests. With n=6, intervals are wide; claims emphasize reproducible mean improvements and architecture honesty rather than universal paired dominance.

F. Deep tabular and true GNN baselines

Table X compares TabNet and true GraphSAGE with the historical GNN proxy and RF. On T1, TabNet (0.049) and true GraphSAGE (0.015) trail RF (0.076) and ECN-v3 (0.115). The historical GNN proxy (0.034) must not be read as message-passing performance. On T2, both deep models also trail telem logistic (0.038).

TABLE IV
MULTI-SEED MEAN T1/T2 RANKING METRICS (SIX INSTANCES; 95% CI). ECN-V3 (T1) IS THE FINAL ANCHORED HEAD; RECOMMENDED T2 HEAD IS LOGISTIC (TELEM).

Method	T1 AUPRC	T1 ROC-AUC	T2 AUPRC	T2 ROC-AUC
ECN-v3 (T1)	**0.115** [0.068, 0.168]	**0.753**	0.015 [0.004, 0.026]	0.637
TabNet	0.049 [0.021, 0.077]	0.630	0.011 [0.004, 0.018]	0.615
GraphSAGE (true)	0.015 [0.006, 0.025]	0.565	0.006 [0.006, 0.007]	0.534
XGBoost	0.047 [0.023, 0.071]	0.701	0.011 [0.007, 0.016]	0.604
CatBoost	0.053 [0.017, 0.088]	0.670	0.012 [0.005, 0.018]	0.571
Gradient boosting	0.073 [0.050, 0.096]	0.759	0.013 [0.008, 0.018]	0.708
Balanced RF	0.047 [0.019, 0.074]	0.775	0.018 [0.005, 0.030]	0.739
Random forest	0.076 [0.043, 0.109]	0.766	0.018 [0.010, 0.025]	0.708
Logistic	0.063 [0.023, 0.103]	0.726	0.038 [−0.003, 0.079]	0.689
LightGBM	0.057 [0.026, 0.088]	0.680	0.021 [−0.004, 0.047]	0.548
Isolation Forest	0.057 [0.025, 0.090]	0.672	0.022 [−0.004, 0.047]	0.646
EWMA	0.040 [0.013, 0.067]	0.548	0.007 [0.005, 0.009]	0.448
GNN proxy	0.034 [0.017, 0.050]	0.685	0.011 [0.007, 0.014]	0.596
MLP sequence	0.008 [0.005, 0.012]	0.362	0.005 [0.004, 0.007]	0.380
Majority	0.010 [0.008, 0.011]	0.500	0.006 [0.005, 0.006]	0.500

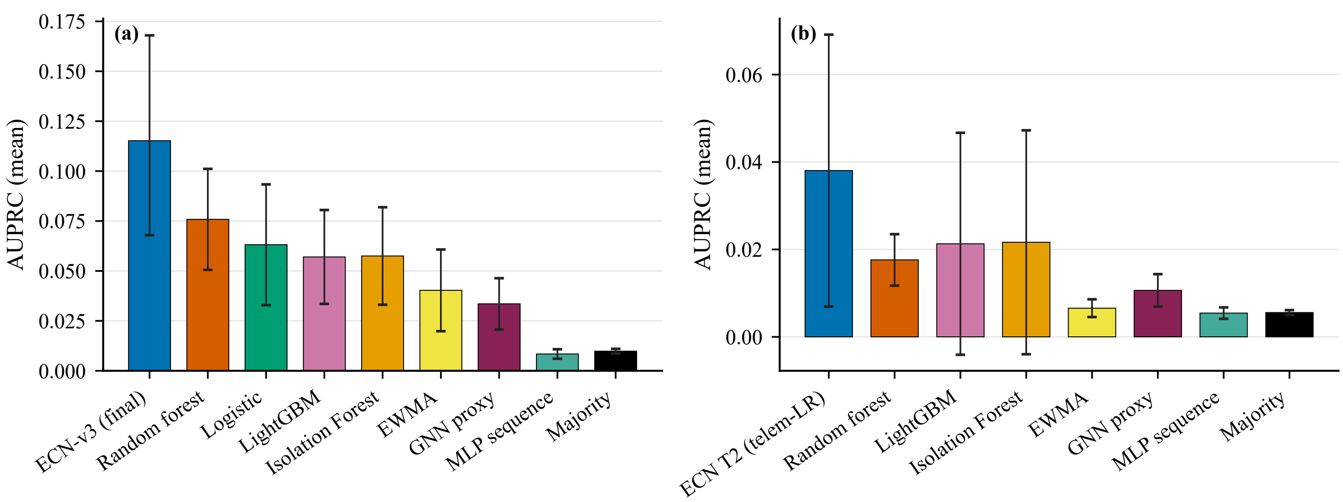

Fig. 3. Multi-seed mean AUPRC with 95% CI. (a) T1 (ECN-v3 = final anchored head). (b) T2 (ECN column = recommended telem-logistic head).

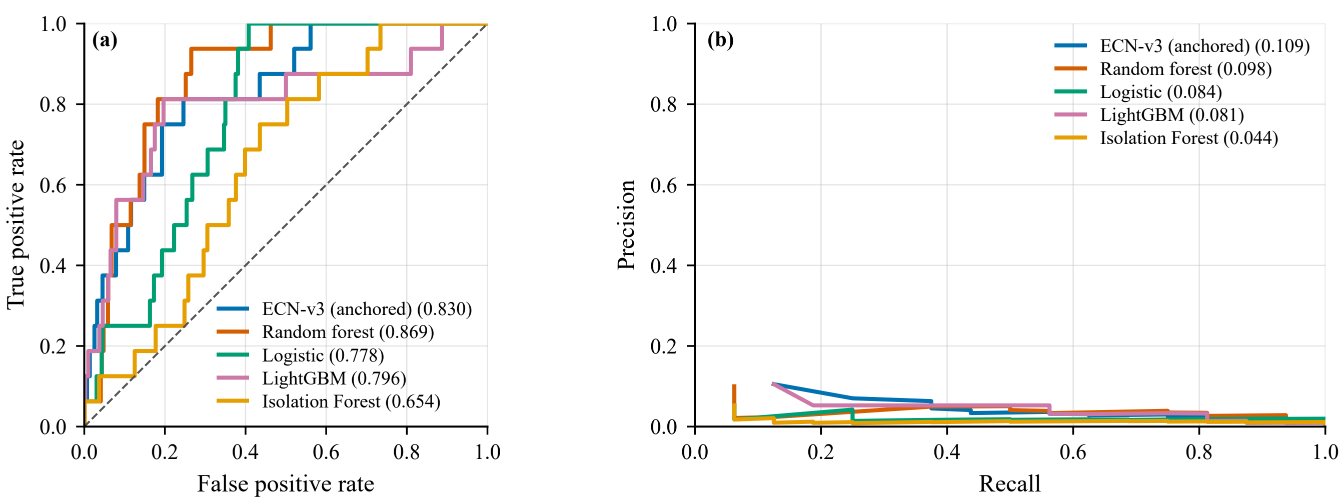

Fig. 4. T1 ROC (a) and precision–recall (b) on the primary frozen instance, including the final anchored ECN-v3 head. Legend values are instance-level ROC-AUC / average precision.

VII. DISCUSSION

A. What to claim

The defensible claims are architectural and methodological: (1) ECNetBench provides a leakage-aware multi-task enter-

TABLE V
PAIRED WILCOXON TESTS AND CLIFF'S δ ON PER-SEED T1/T2 AUPRC (n=6).

Task	Baseline	p	Cliff δ	Proposed	Baseline
T1	TabNet	0.062	+0.667	0.115	0.049
	GraphSAGE	0.031	+0.944	0.115	0.015
	XGBoost	0.094	+0.722	0.115	0.047
	CatBoost	0.156	+0.611	0.115	0.053
	LightGBM	0.062	+0.611	0.115	0.057
	Gradient boosting	0.219	+0.500	0.115	0.073
	Balanced RF	0.031	+0.667	0.115	0.047
	Random forest	0.688	+0.389	0.115	0.076
	Logistic	0.031	+0.556	0.115	0.063
	Isolation Forest	0.031	+0.667	0.115	0.057
	EWMA	0.062	+0.778	0.115	0.040
	Threshold	0.031	+0.778	0.115	0.032
	MLP sequence	0.031	+1.000	0.115	0.008
	GNN proxy	0.062	+0.778	0.115	0.034
	Majority	0.031	+1.000	0.115	0.010
T2	Logistic	0.844	−0.500	0.015	0.038
	Random forest	0.688	−0.333	0.015	0.018

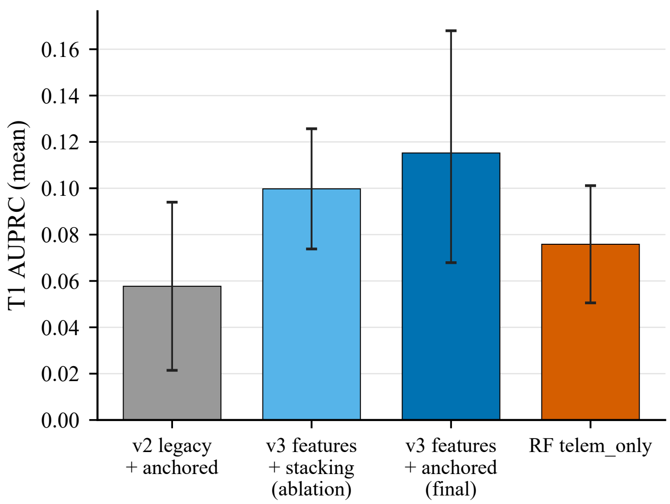

Fig. 5. T1 architecture selection over six seeds (mean AUPRC with 95% CI).

TABLE VI
ABLATION MEAN AUPRC ON T1/T2 (SIX SEEDS, 95% CI).

Task	Ablation	Mean	95% CI
T1	full	0.111	[0.042, 0.180]
	no_twin	0.115	[0.043, 0.187]
	no_nbr	0.116	[0.051, 0.181]
	telem_only	0.069	[0.014, 0.123]
	twin_only	0.028	[0.013, 0.044]
T2	full	0.015	[0.004, 0.026]
	no_twin	0.014	[0.003, 0.025]
	no_nbr	0.014	[0.003, 0.025]
	telem_only	0.036	[−0.007, 0.078]
	twin_only	0.016	[0.008, 0.023]

TABLE VII
MODULE AUPRC CONTRIBUTION (full − ablated) FROM FULL-SUITE ABLATIONS.

Task	Telemetry	Neighbor	Digital Twin
T1	0.082	−0.005	−0.004
T2	−0.000	0.001	0.001

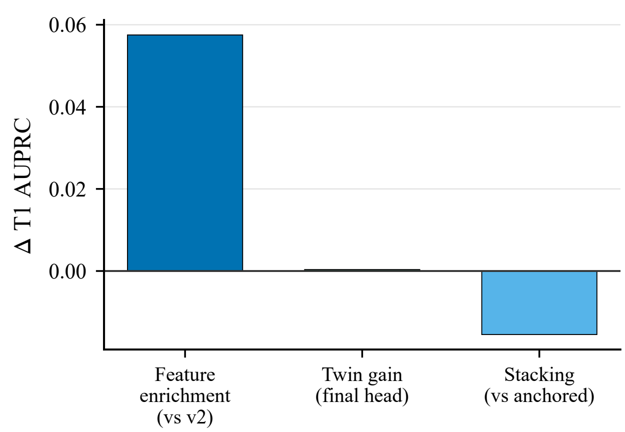

Fig. 6. T1 AUPRC deltas from architecture selection: feature enrichment, twin gain, and stacking versus anchored fusion.

TABLE VIII
CALIBRATION QUALITY (BRIER SCORE; LOWER IS BETTER).

Task	Method	Mean	95% CI
T1	ECN-v3 (selection)	0.151	[0.093, 0.209]
T1	Random forest	0.023	[0.011, 0.034]
T1	Logistic	0.233	[0.174, 0.292]
T2	ECN proposed	0.209	[0.119, 0.299]
T2	Logistic	0.258	[0.208, 0.307]
T2	Random forest	0.012	[0.009, 0.016]

prise benchmark with multi-seed frozen instances; (2) ECN-v3 closes the loop from detection to healing *decision support* with TreeSHAP RCA; (3) leakage-safe temporal enrichment is the primary driver of the T1 lift from v2 to the final head; (4) anchored fusion is preferred over stacking on this benchmark (higher mean AUPRC, lower complexity); (5) T2 remains best served by a telem logistic head under extreme rarity.

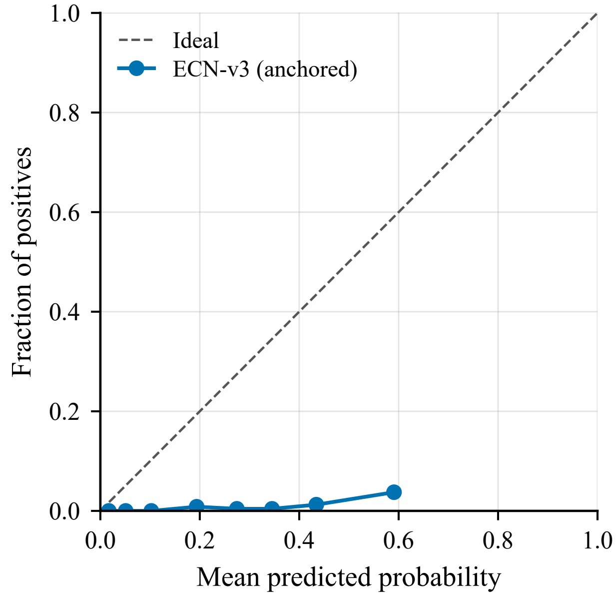

Fig. 7. T1 reliability diagram on the primary frozen instance for the anchored ECN-v3 head.

TABLE IX
RCA MACRO-F1, HEALING SCORES, AND EVALUATION WALL-CLOCK COST (SIX SEEDS).

Metric	Setting	Mean [95% CI]
T3 RCA macro-F1	proposed	0.254 [0.090, 0.418]
Healing macro-F1	with RCA category	1.000 [1.000, 1.000]
Healing macro-F1	without RCA category	0.190 [0.099, 0.282]
Eval. wall time (s/seed)	full suite	119.7 [111.8, 127.6]

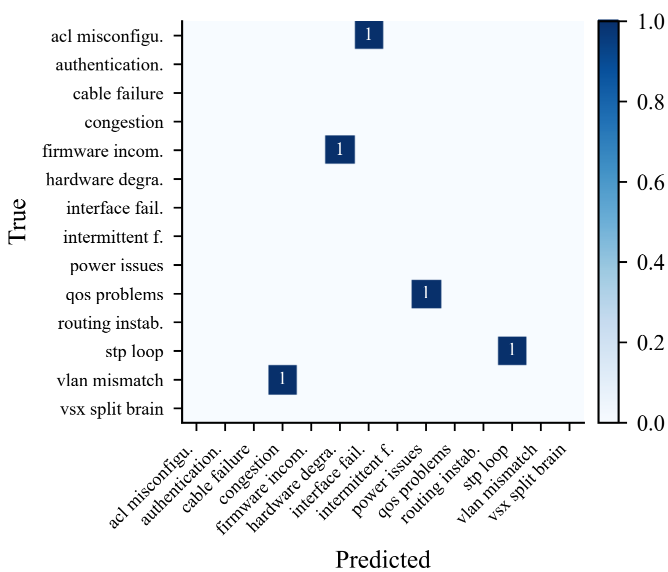

Fig. 8. Representative T3 RCA confusion matrix (RF+TreeSHAP) on the primary frozen instance.

B. What not to claim

We do not claim statistically significant superiority over random forest on T1 (p=0.688), significant superiority of anchored over stacking (p=0.375), live deployment on physical AOS-CX switches, or that multi-agent fusion alone beats strong tabular detectors without feature enrichment. TabNet and true GraphSAGE likewise do not overturn the evidence-selected heads on this protocol. Impact AUPRC of 1.0 for some models on this synthetic label construction is reported for completeness but interpreted cautiously as potential label/feature collinearity rather than solved service-impact prediction in the wild.

TABLE X
DEEP BASELINES VERSUS ECN-V3 AND RF ON T1/T2 (SIX-SEED MEANS, 95% CI).

Method	T1 AUPRC	T2 AUPRC
ECN-v3 final	0.115 [0.068, 0.168]	—
TabNet	0.049 [0.021, 0.077]	0.011 [0.004, 0.018]
GraphSAGE (true)	0.015 [0.006, 0.025]	0.006 [0.006, 0.007]
GNN proxy (LightGBM)	0.034 [0.017, 0.050]	0.011 [0.007, 0.014]
Random forest (telem)	0.076 [0.043, 0.109]	0.018 [0.010, 0.025]

C. Practical NetOps implications

Operators can use ECN as a sandbox for ranking alerts, proposing RCA categories with SHAP attributions, and suggesting remediation classes before human approval. Prefer the anchored head for T1 triage; prefer telem logistic scores for T2 horizon ranking; apply optional Beta calibration when displaying probabilities.

D. Governance and operational implications

Because ECN-v3 closes a multi-agent loop that can recommend remediation classes, deployment should retain human approval gates for delegated actions, consistent with governance analyses of agentic systems and agent-to-agent delegation [21], [22], [28]. Leakage-safe temporal evaluation and checksummed instances support operational-data auditability [24], while cybersecurity risk-control and zero-exposure guidance caution against exposing privileged recovery pathways through automated tooling [23], [25]. If generative or retrieval-augmented assistants are later attached to NetOps workflows, enterprise generative-AI security and agentic-AI governance frameworks provide complementary controls beyond detector metrics [26], [27]. Semantic tracing practices for LLM-based multi-agent stacks offer a useful template for governing agent decisions alongside TreeSHAP RCA [29]. Digital-transformation studies further remind operators that legacy constraints and human resistance often dominate outcomes even when models improve ranking scores [4]–[7].

VIII. LIMITATIONS AND THREATS TO VALIDITY

Table XI summarizes the principal threats to validity. Construct validity risks include synthetic incident generators that may over-structure causal chains. Internal validity is strengthened by frozen instances, checksums, and relative-path reproduction. Conclusion validity is limited by n=6 and wide confidence intervals. External validity requires future evaluation on anonymized production telemetry.

TABLE XI
THREATS TO VALIDITY AND CORRESPONDING MITIGATIONS.

Threat	Mitigation / residual risk
Synthetic data	Realism audits; still not a substitute for production traces.
External validity	AOS-CX-style assumptions; other vendors/fabrics may differ.
Label leakage	Temporal freeze and feature audits; healing full vs. `no_rca_cat`.
Small n seeds	Six seeds with confidence intervals; tests remain underpowered.
Topology scale	19 devices / 31 links; scalability to large fabrics unproven.
Healing semantics	Decision support only; no live actuation evidence.

IX. CONCLUSION

We presented ECNetBench and ECN-v3, a digital-twin multi-agent framework evaluated under leakage-safe multi-seed protocols. The final anchored T1 head attains mean AUPRC 0.115, above random forest (0.076) and the prior v2 configuration (0.058), while stacking is a negative ablation and paired tests versus RF remain non-significant at n=6. TabNet and true GraphSAGE do not overturn these selections. The recommended T2 head is telemetry logistic regression. The primary scientific value is a reproducible benchmark plus a complete cognitive NetOps pipeline with evidence-selected fusion and honest statistics—not a universal detector champion. Future work includes larger topologies, production-trace calibration, richer temporal GNNs, human-in-the-loop healing studies, and stronger governance-oriented observability under enterprise AI security controls [26]–[29].

APPENDIX A
SUPPLEMENTARY NOTES

A. Reproduction commands

```
python -m venv .venv
pip install -r requirements.txt
python scripts/download_instances.py
python scripts/verify_instances.py
pytest -q
python evaluation/trace_t1_gain.py
python evaluation/select_t1_architecture.py
python evaluation/run_full_evaluation.py
python paper/overleaf/scripts/\
regenerate_pub_figures_v3.py
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

B. Traceability

Final T1 numbers map to `results/manuscript_ready_numbers.json` and `results/v3_gated/t1_architecture_selection.json`. Baseline/T2/RCA aggregates map to `results/aggregate_v3.json` and `results/per_seed/*.json`. Additional reports: `reports/FINAL_ARCHITECTURE_SELECTION.md`, `reports/T1_IMPROVEMENT_TRACEABILITY.md`. The Overleaf project is synchronized under `paper/overleaf/`.

C. Additional figures

Per-seed ROC/PR, confusion matrices, and calibration plots for T1/T2/T3 are included under `figures/` as vector PDF and 600 dpi PNG pairs. The primary-instance T2 curves and T1 confusion matrix appear on the following page; remaining seeds are archived for completeness.

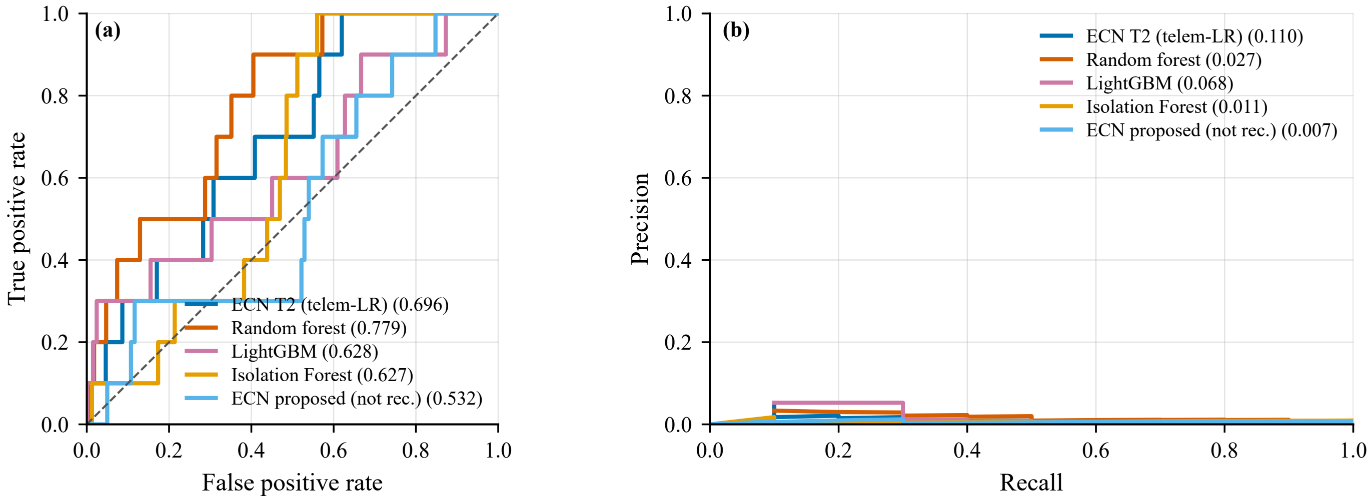

Fig. 9. T2 ROC (a) and precision–recall (b) curves on frozen instance v1.1.0-INST. The recommended T2 operating head is telemetry logistic regression (labeled ECN T2 (telem-LR)); the multi-agent proposed T2 curve is shown only for contrast.

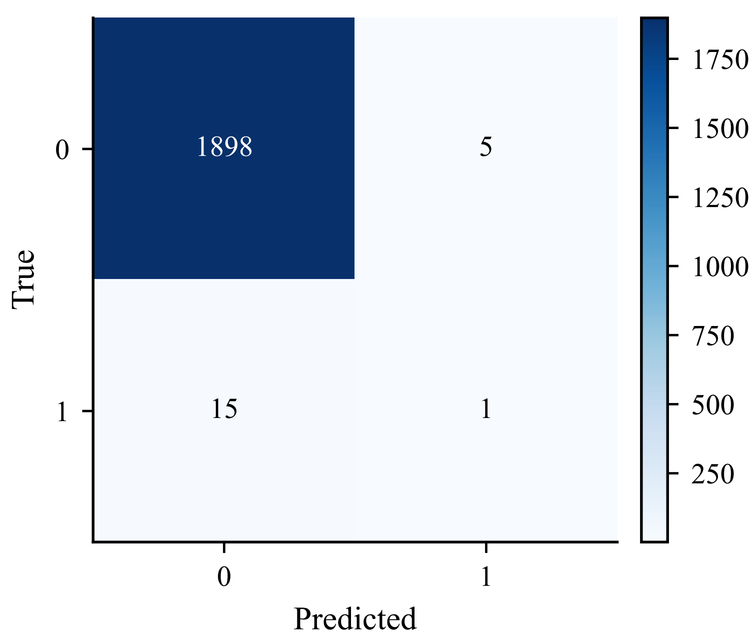

Fig. 10. T1 confusion matrix on instance v1.1.0-INST at a validation-tuned threshold (threshold unused for AUPRC).

Data Availability

Frozen ECNetBench SQLite instances and checksums will be released with the camera-ready version of this work. For double-blind review, instances are provided via an anonymized supplementary archive whose contents match relative path `benchmark/instances/` and checksums in `benchmark/INSTANCE_CHECKSUMS.json` (SHA-256 digests; reviewers should verify the archive against that manifest). Evaluation scripts read instances only through relative paths.

Code Availability

Source code for the generator, Digital Twin, agents, baselines, ablations, and evaluation harness will be released under an MIT license with the camera-ready version. An anonymized code archive matching the submitted artifact hash is provided for review; public repository URLs are withheld during double-blind review and restored upon acceptance. Exact reproduction commands appear in Appendix A and the companion reproducibility notes.

Author Contributions

Conceptualization, methodology, software, validation, investigation, data curation, writing—original draft, and visualization were performed by the author(s). Formal analysis and statistical testing follow the published evaluation scripts.

Conflict of Interest

The authors declare no competing interests.

Acknowledgments

[Acknowledgments to be completed upon acceptance.]

References

[1] Y. Dang, Q. Lin, and P. Huang, "AIOps: Real-world challenges and research innovations," *IEEE/ACM International Conference on Software Engineering: Companion Proceedings (ICSE-Companion)*, pp. 4–5, 2019.

[2] P. Notaro, J. Cardoso, and M. Gerndt, "A survey of AIOps methods for failure management," *ACM Transactions on Intelligent Systems and Technology*, vol. 12, no. 6, pp. 1–45, 2021.

[3] P. Almasan, M. Ferriol-Galmés, J. Paillisse, J. Suárez-Varela, D. Perino, D. López, A. A. P. Perales, P. Harvey, L. Ciavaglia, L. Wong, V. Ramakrishna, S. Xiao, X. Shi, X. Cheng, A. Cabellos-Aparicio, and P. Barlet-Ros, "Network digital twin: Context, enabling technologies, and opportunities," *IEEE Communications Magazine*, vol. 60, no. 11, pp. 22–27, 2022.

[4] A. Rah, "Digital transformation beyond technology: Governance, human resistance, and cybersecurity," Preprints.org, 2026, preprint.

[5] ——, "Why digital transformation fails: Human resistance, legacy systems, and IT modernization challenges," 2026, preprint.

[6] ——, "Governance human resistance in digital transformation," 2026, preprint.

[7] ——, "A governance-oriented analytical research paper on digital transformation and cybersecurity," 2026, preprint.

[8] S. Knight, H. X. Nguyen, N. Falkner, R. Bowden, and M. Roughan, "The Internet Topology Zoo," *IEEE Journal on Selected Areas in Communications*, vol. 29, no. 9, pp. 1765–1775, 2011.

[9] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," in *Military Communications and Information Systems Conference (MilCIS)*, 2015, pp. 1–6.

[10] *ITU-T Y.3090: Digital Twin Network – Requirements and Architecture*, International Telecommunication Union, 2022.

[11] C. Zhou, H. Yang, X. Duan, D. Lopez, A. Pastor, Q. Wu, M. Boucadair, and C. Jacquenet, "Digital twin network: Concepts and reference architecture," IETF Internet-Draft draft-irtf-nmrg-network-digital-twin-arch, 2024, work in Progress.

[12] ETSI, "AI in the evolution of autonomous networks," European Telecommunications Standards Institute, White Paper 69, 2024.

[13] V. Chandola, A. Banerjee, and V. Kumar, "Anomaly detection: A survey," *ACM Computing Surveys*, vol. 41, no. 3, pp. 1–58, 2009.

[14] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.

[15] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[16] K. Rusek, J. Suárez-Varela, A. Mestres, P. Barlet-Ros, and A. Cabellos-Aparicio, "Unveiling the potential of graph neural networks for network modeling and optimization in SDN," *Proceedings of the ACM Symposium on SDN Research (SOSR)*, pp. 140–151, 2019.

[17] S. O. Arik and T. Pfister, "TabNet: Attentive interpretable tabular learning," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 8, 2021, pp. 6679–6687.

[18] S. Orlowski, R. Wessäly, M. Pióro, and A. Tomaszewski, "SNDlib 1.0—survivable network design library," *Networks*, vol. 55, no. 3, pp. 276–286, 2010.

[19] A. Clemm, L. Ciavaglia, L. Z. Granville, and J. Tantsura, "Intent-based networking—concepts and definitions," *IETF RFC 9315*, 2022.

[20] M. Wooldridge, *An Introduction to MultiAgent Systems*, 2nd ed. John Wiley & Sons, 2009.

[21] A. Rah, "Agentic artificial intelligence: A systematic review of architectures, benchmarks, and governance frameworks," 2026, preprint.

[22] ——, "Agentic and multi-agent systems: A systematic review of tool use, benchmarks, and governance," 2026, preprint.

[23] ——, "Cybersecurity risk controls in IT organizations," 2026, preprint.

[24] ——, "Auditing operational data records to improve trust and risk assessments," 2026, preprint.

[25] ——, "Reducing insider risk in identity recovery through zero-exposure governance," 2026, preprint.

[26] ——, "Enterprise generative AI: A systematic review of security risks and governance," 2026, preprint.

[27] A. Rah and S. Hahues, "Retrieval-augmented generation (RAG), generative AI, and agentic AI governance: An integrated enterprise governance prioritization architecture," 2026, preprint.

[28] A. Rah, "Agent-to-agent business ecosystems: Delegated commercial agency, governance, and a reference architecture for autonomous enterprise commerce," 2026, preprint.

[29] ——, "Semantic tracing in LLM-based multi-agent systems using LangChain, LangGraph, and LangSmith for AI governance," 2026, preprint.